National Testing Agency

Question Paper Name :	B Tech 5th Apr 2026 Shift 1
Subject Name :	B. Tech
Creation Date :	2026-04-05 14:04:32
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	6952785
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	69527825
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527825
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 695278301 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $a, b \in \mathbb{C}$. Let α, β be the roots of the equation $x^2 + ax + b = 0$. If $\beta - \alpha = \sqrt{11}$ and $\beta^2 - \alpha^2 = 3i\sqrt{11}$, then $(\beta^3 - \alpha^3)^2$ is equal to:

Options :

6952781021. 160

6952781022. 176

6952781023. 194

6952781024. 187

Question Number : 2 Question Id : 695278302 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the sum of the first n terms of an A.P. be $3n^2 + 5n$. Then the sum of squares of the first 10 terms of the A.P. is:

Options :

6952781025. 10220

6952781026. 12860

6952781027. 15220

6952781028. 19780

Question Number : 3 Question Id : 695278303 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let A be a 3×3 matrix such that

$$A^T\begin{bmatrix}1\\0\\1\end{bmatrix}=\begin{bmatrix}5\\2\\2\end{bmatrix}, A^T\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}3\\1\\1\end{bmatrix}, A\begin{bmatrix}1\\0\\1\end{bmatrix}=\begin{bmatrix}3\\4\\4\end{bmatrix} \text{ and } A\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}1\\3\\1\end{bmatrix}.$$

If $\det(A) = 1$, then $\det(\text{adj}(A^2 + A))$ is equal to:

Options :

6952781029. 16

6952781030. 25

6952781031. 49

6952781032. 64

Question Number : 4 Question Id : 695278304 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the system of linear equations in x, y, z:
$x + 2y + tz = 0$,
$6x + y + 5tz = 0$,
$3x + t^2y + f(t)z = 0$,
where $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function. If this system has infinitely many solutions for all $t \in \mathbb{R}$, then f

Options :

6952781033. is a constant function

6952781034. is strictly increasing on $\mathbb{R}$

6952781035. is strictly decreasing on $\mathbb{R}$

6952781036. has two critical points

Question Number : 5 Question Id : 695278305 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$\sum_{n=1}^{10}\left(\frac{528}{n(n+1)(n+2)}\right)$ is equal to:

Options :

6952781037. 65

6952781038. 130

6952781039. 220

6952781040. 440

Question Number : 6 Question Id : 695278306 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\tan A$, $\tan B$, where $A, B \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2\left(\frac{A+B}{2}\right)$ is equal to:

Options :

6952781041. $10+\sqrt{10}$

6952781042. $10-2\sqrt{10}$

6952781043. $10-3\sqrt{10}$

6952781044. $10-\sqrt{10}$

Question Number : 7 Question Id : 695278307 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the envelope just two consecutive letters AN are visible. The probability, that the letter came from ANANTPUR, is:

Options :

6952781045. $\frac{7}{10}$

6952781046. $\frac{10}{17}$

6952781047. $\frac{12}{19}$

6952781048. $\frac{7}{19}$

Question Number : 8 Question Id : 695278308 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The mean deviation about the mean for the data

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

is equal to:

Options :

6952781049. $\frac{40}{13}$

6952781050. $\frac{42}{13}$

6952781051. $\frac{44}{13}$

6952781052. $\frac{46}{13}$

Question Number : 9 Question Id : 695278309 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let a focus of the ellipse $E: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ be S(4, 0) and its eccentricity be $\frac{4}{5}$. If the point P(3, α) lies on E and O is the origin, then the area of ΔPOS is equal to:

Options :

6952781053. 12/5

6952781054. 14/5

6952781055. 24/5

6952781056. 48/5

Question Number : 10 Question Id : 695278310 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let P be a moving point on the circle $x^2 + y^2 - 6x - 8y + 21 = 0$. Then, the maximum distance of P from the vertex of the parabola $x^2 + 6x + y + 13 = 0$ is equal to:

Options :

6952781057. 8

6952781058. 10

6952781059. 12

6952781060. 9

Question Number : 11 Question Id : 695278311 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In an equilateral triangle PQR, let the vertex P be at (3, 5) and the side QR be along the line $x + y = 4$. If the orthocentre of the triangle PQR is (α, β), then $9(\alpha + \beta)$ is equal to:

Options :

6952781061. 16

6952781062. 27

6952781063. 36

6952781064. 48

Question Number : 12 Question Id : 695278312 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The sum of all the integral values of p such that the equation
$3\sin^2 x + 12\cos x - 3 = p,\ x \in \mathbb{R}$, has at least one solution, is:

Options :

6952781065. -54

6952781066. -60

6952781067. -75

6952781068. -84

Question Number : 13 Question Id : 695278313 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The square of the distance of the point P(5, 6, 7) from the line
$\frac{x-2}{2} = \frac{y-5}{3} = \frac{z-2}{4}$ is equal to:

Options :

6952781069. 3

6952781070. 5

6952781071. 6

6952781072. 8

Question Number : 14 Question Id : 695278314 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\vec{a} = \sqrt{7}\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + 2\hat{k}$. If $\vec{r}$ is a vector such that
$\vec{r} \times \vec{a} + \vec{a} \times \vec{b} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$, then $|3\vec{r}|^2$ is equal to:

Options :

6952781073. 44

6952781074. 54

6952781075. 86

6952781076. 132

Question Number : 15 Question Id : 695278315 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The square of the distance of the point of intersection of the lines
$\vec{r} = \left(\hat{i} + \hat{j} - \hat{k}\right) + \lambda\left(a\hat{i} - \hat{j}\right),\ a \neq 0$ and $\vec{r} = \left(4\hat{i} - \hat{k}\right) + \mu\left(2\hat{i} + a\hat{k}\right)$ from the origin is:

Options :

6952781077. 5

6952781078. 10

6952781079. 17

6952781080. 26

Question Number : 16 Question Id : 695278316 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The area of the region $R = \{(x, y): xy \le 27, 1 \le y \le x^2\}$ is equal to:

Options :

6952781081. $78\log_e 3 - \frac{52}{3}$

6952781082. $54\log_e 3 - \frac{52}{3}$

6952781083. $54\log_e 3 - \frac{26}{3}$

6952781084. $54\log_e 3 + \frac{26}{3}$

Question Number : 17 Question Id : 695278317 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The product of all possible values of α, for which

$$\lim_{x\to 0}\left(\frac{1-\cos(\alpha x)\cos\left((\alpha+1)x\right)\cos\left((\alpha+2)x\right)}{\sin^2\left((\alpha+1)x\right)}\right) = 2,$$ is:

Options :

6952781085. -2

6952781086. 1

6952781087. -1

6952781088. $\frac{5}{4}$

Question Number : 18 Question Id : 695278318 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of the integral $\int_0^{\infty} \frac{\log_e(x)}{x^2+4}\,dx$ is:

Options :

6952781089. $\frac{\pi\log_e(2)}{2}$

6952781090. $\frac{\pi\log_e(2)}{4}$

6952781091. $1+\pi\log_e(2)$

6952781092. $2+\pi\log_e(2)$

Question Number : 19 Question Id : 695278319 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function such that $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3}$ for all $x, y\in\mathbb{R}$, and $f'(0)=3$. Then the minimum value of the function $g(x)=3+e^x f(x)$, is:

Options :

6952781093. $3\left(\frac{e+1}{e}\right)$

6952781094. $3\left(\frac{e-1}{e}\right)$

6952781095. $\frac{3-e}{e}$

6952781096. $3e$

Question Number : 20 Question Id : 695278320 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of the integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\left(\frac{4-\operatorname{cosec}^2 x}{\cos^4 x}\right)dx$ is:

Options :

6952781097. $\frac{11}{\sqrt{3}}$

6952781098. $\frac{16}{\sqrt{3}}$

6952781099. $\frac{32}{3\sqrt{3}}$

6952781100. $\frac{64}{3\sqrt{3}}$

Mathematics Section B

Section Id :	69527826
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5

Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527826
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 695278321 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let A = {1, 2, 3, 4, 5, 6}. The number of one-one functions $f: A \to A$ such that $f(1) \geq 3, f(3) \leq 4$ and $f(2) + f(3) = 5$, is_____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 22 Question Id : 695278322 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Two players A and B play a series of games of badminton. The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is_____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 695278323 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the sum of the coefficients of x^7 and x^{14} in the expansion of $\left(\frac{1}{x^3} - x^4\right)^n$, $x \neq 0$, is zero, then the value of n is _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 695278324 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $\frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^{p-1}}{1+2^{2p-1}}\right) = \alpha$, then $\tan\alpha$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Possible Answers :

1

Question Number : 25 Question Id : 695278325 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $y = y(x)$ be the solution of the differential equation $x\sin\left(\frac{y}{x}\right)dy = \left(y\sin\left(\frac{y}{x}\right) - x\right)dx$, $y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{y\left(e^{12}\right)}{e^{12}}\right)$. Then the number of integral value of p, for which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $r \leq 6$, is_____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Physics Section A

Section Id :	69527827
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527827
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 695278326 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7th Vernier division coincides with a main scale reading. If the value of 1 main scale division is 1 mm and there are 10 Vernier scale divisions, then the Vernier caliper has

Options :

6952781106. 0.07 cm negative zero error

6952781107. 0.7 cm negative zero error

6952781108. 0.07 cm positive zero error

6952781109. 0.7 cm positive zero error

Question Number : 27 Question Id : 695278327 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

L, C and R represents physical quantities inductance, capacitance and resistance respectively. The dimensional formula $M L^2 T^{-4} A^{-2}$ corresponds to _____.